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Management and analysis of high-throughput sequence data for infectious animal diseases

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Summary

Advances in technology and decreasing costs have accelerated the use of high-throughput sequencing (HTS) for both diagnosis and characterisation of infectious animal diseases. High-throughput sequencing offers several advantages over previous techniques, including rapid turnaround times and the ability to resolve single nucleotide changes among samples, both of which are important for epidemiological investigations of outbreaks. However, due to the plethora of genetic data being routinely generated, the storage and analysis of these data are proving challenging in their own right.

In this article, the authors provide insight into the aspects of data management and analysis that should be considered before adopting HTS for routine animal health diagnostics. These elements fall largely into three interrelated categories: data storage, data analysis and quality assurance. Each has numerous complexities and may need to be adapted as HTS evolves. Making appropriate strategic decisions about bioinformatic sequence analysis early on in project development will help to avert major issues in the long term.

Keywords

Bioinformatics – Data strategy – High-throughput sequence data – Infectious animal disease characterisation – Infectious animal disease diagnosis – Quality assurance.

Introduction

With the ongoing developments in instruments for high-throughput sequencing (HTS), genetic sequence data can be generated in ever-increasing quantities for diminishing costs and with shorter turnaround times. This has been driven primarily by the ability to improve sensitivity and characterise multiple DNA fragments in parallel. Because of these advances, HTS is commonly used in most fields of biology and has led to significant advances in the understanding of evolution, microbiology and medicine.

In the context of animal health, HTS is increasingly being used for both the diagnosis and characterisation of infectious animal diseases [1]. Whilst HTS was first used for disease outbreak investigations, it is now being applied to more routine surveillance programmes. For example, the value of HTS was demonstrated when concurrent cases of H5N8 avian influenza were detected across Europe in November 2014. Whole genome sequence comparisons showed a high degree of sequence similarity between samples from Germany, the Netherlands and Asia, suggesting a common source [2] and further demonstrating the role of migratory birds in transmission [3]. Since then, it has become commonplace for laboratories to undertake HTS of viral samples, adding them to databases for surveillance purposes and to facilitate tracking of emerging pathogen variants (e.g. https://www.gisaid.org/collaborations/global-collaboration-on-h5n8).

Similarly, HTS and whole genome sequencing are being used for outbreak investigation and surveillance of bacterial pathogens such as *Mycobacterium bovis* [4] and *Salmonella* spp. [5]. However, due to the flexibility and non-selectivity of this approach (compared to other nucleic acid tests), HTS is suitable for a wider selection of applications including, for example, the simultaneous detection of multiple pathogens, screening for unknown disease-causing agents, or a single universal diagnostic assay [1]. With new sequencing technologies being developed, it is likely that uses of HTS will be extended to other challenges, such as characterising host responses to infection via full-length RNA transcript sequencing.

Consequently, an ever-increasing quantity of HTS data is being generated and analysed to support animal health but, without careful planning, the volume of data can soon become overwhelming. Therefore, careful consideration of how data will be managed and analysed is required before embarking on a study. Here, the authors discuss practical considerations when managing HTS data and how to choose the most appropriate analyses. Such considerations can be broken down primarily into three areas: data storage (where, and for how long); analysis (processing to attain the required outcomes); and quality assurance (understanding performance criteria) (Figure 1).

While this overview may not be exhaustive, the authors hope that it will encourage researchers to plan for these issues before starting a study involving HTS, or setting up a facility to provide HTS for multiple researchers, diagnosticians or veterinarians. Table I contains a glossary of terms used throughout this article.

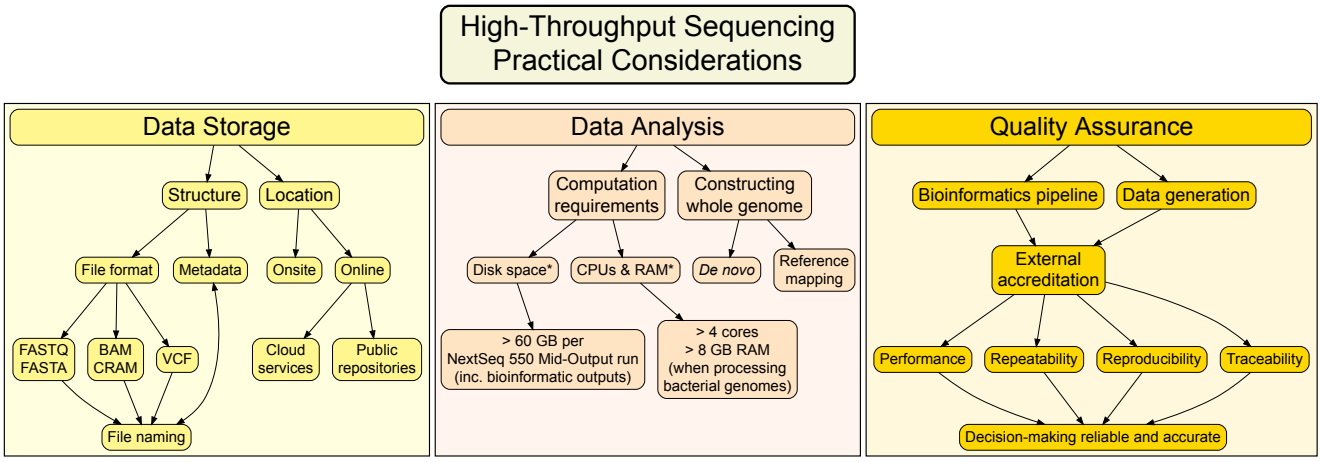
Practical considerations

Data storage

The considerable amount of data generated by HTS is a major challenge when adopting it for routine disease surveillance. For example, Illumina's mid-output instrument can generate 30 gigabytes (GB) of data daily, which equates to a few hundred bacterial isolates, or around ten samples for viral genomes directly sequenced from tissue samples. Processed data files from analytical pipelines are likely to double the volume of data to be held. After just a handful of sequencing runs, the volume of data adds up, so thought should be given as to how these data are to be handled, where they should be stored, and for how long.

Firstly, the organisation of the data should be determined – that is, a data structure should be defined. High-throughput sequencing instruments produce data in batches, determined by the number and type of samples multiplexed on a sequencing run, and individual output data files are unique to individual biological samples. The biological sample is associated with metadata, such as source details (host species, temporal and geographic details and, potentially, clinical signs related to the disease). Retaining a link between the metadata and the HTS data files is essential for downstream interpretation and, in the authors' experience, is best achieved by using a Laboratory Information Management System (LIMS). Labelling samples 'Sample 1', 'Sample 2', etc. is not informative and likely to cause confusion sooner rather than later. Linking a unique identifier to additional metadata stored in the linked database (or LIMS) will facilitate tracking.

However, the HTS files must also be organised into folders or directories, according to strict criteria. A flat storage approach in which a single folder contains a list of files with unique identifiers will be difficult to manage. The simplest approach would be to label folders by the date of sequencing but, to facilitate downstream processing, it would then be more prudent to group them by source material, pathogen or analysis required. For the purposes of traceability, being able to link data for a given sample back to the laboratory, and



BAM: binary alignment map
CPUs: central processing units
CRAM: compressed reference-oriented alignment map
FASTA: plain text format for storing biological sequences which can be compressed
FASTQ: plain text format for storing biological sequences which can be compressed
GB: gigabyte
RAM: random access memory
VCF: variant call format

Figure 1

Flow chart highlighting practical considerations for planning a high-throughput sequencing study

This chart shows the practical issues that should be incorporated into strategic planning for a new high-throughput sequencing study for animal pathogen surveillance, before implementation. The three key areas to consider are: data management, data analysis, and quality assurance

*Note that storage space and CPUs/RAM will depend on the genome size of the target organism and the type of analysis being performed

Table I

Glossary

Term	Description
BAM	Binary alignment map – a binary version of the sequence alignment/map file format, which contains information about the position and similarity of sequencing reads to a reference genome
CPU	Central processing unit – the key part of a computer that runs programs and analysis. CPUs can have multiple cores that allow processes to be run in parallel, thus decreasing the time taken to complete the computation
CRAM	Compressed reference-oriented alignment map – a compressed file format for storing biological sequences aligned to a reference sequence
De novo assembly	A whole genome is assembled from raw reads, without a reference genome, by piecing together the reads based on overlaps
DNA sequencing	A technique for determining the exact sequence of nucleotides, or bases, in a DNA molecule
FASTA	A plain text file format for storing DNA sequences, such as genome assemblies and consensus sequence alignments. A FASTA file only stores a sequence ID and the DNA sequence
FASTQ	A plain text file format for storing raw reads output from high-throughput sequencing. A FASTQ file stores a sequence ID, the DNA sequence, and quality scores that record the accuracy of each base call in the read. Files are typically compressed for storage
HTS	High-throughput sequencing – technologies that sequence DNA or RNA in massive quantities at cost-effective prices and with rapid turnaround times
RAM	Random access memory – computer memory used to store working data. Higher RAM is required for processing the large data files generated by HTS
Read	A partial sequence of a single DNA fragment
Read length	The number of base pairs (bp) sequenced from a DNA fragment. With 'short-read' sequencing technology, read lengths are typically 100–250 bp. Reads from 'long-read' technologies can be up to hundreds of thousands of bases long
Reference mapping	A whole genome for an individual is constructed by aligning (also called mapping) raw reads to a reference genome. The reference is usually a quality genome assembly from the same species (or isolate) or a closely related species if the former is not available
VCF	Variant call format – a text file format for storing variations in DNA sequences between individuals in a population of samples, or between a single sample and a reference genome

even the precise instrument that generated the data, would be beneficial.

Secondly, alongside the data structure, the file type or format should be decided upon. Several community standard file formats have emerged and are preferable to those produced by proprietary software, which generates its own format and can only be read by software developed by the organisation/individual concerned. Standardising data formats allows the integration of data from multiple sources and a wider use of analytical software. The most used format for raw data output from HTS instruments is FASTQ [6], which consists of the nucleotide sequence (as with the FASTA format), but with an additional quality metric that indicates the statistical confidence that each individual base call is correct. These are essentially plain text files for storing biological sequences that can be compressed using standard tools, which is vital when large numbers of files are to be stored.

The next stage should be to determine where to store the data. Laboratories often start with on-site solutions, such as network-attached storage, but then find that there is an ongoing need to maintain such storage, expand the available space and store backups. Alternatives, such as Cloud solutions (e.g. Amazon Web Services, Azure, Google Cloud, etc., with appropriate assurances around data security), are likely to be more flexible; maintenance, expansion and backups are dealt with by the provider and costs are determined by how much space is used. Ultimately, data storage costs are also likely to dictate retention – as more data are amassed the costs will continue to rise and decisions must be made about when old data (probably produced on older sequencing instruments) are no longer valuable. There is an argument that raw data (i.e. FASTQ files) should be retained long term, especially if genome data have been assembled, because continual improvements in assembly software and the potential to combine these data with complementary data from other sources mean that future re-analysis could improve the accuracy and utility of the raw data.

Data can also be retained indefinitely in public repositories, such as those hosted by the European Bioinformatics Institute (EBI) or National Center for Biotechnology Information (NCBI). These repositories will accept either raw data as they are emitted from sequencing instruments (e.g. FASTQ files), or the outputs from analysis (e.g. assembly or mapping – FASTA consensus sequences). These systems are funded by governments to facilitate data sharing so, whilst there is no monetary cost for storage, the data will be available for anyone to view or analyse. There is often an

embargo period to keep data private, but this is finite. Some metadata are required to accompany the sequence data to increase the data's public value, but precise information (e.g. exact sampling location) is not essential and can be less identifiable (e.g. a geographic district). As a solution for long-term data deposition, once the implications of the data are no longer sensitive, public repositories are an ideal solution. Alternatively, more restrictive repositories, which require registration and include a degree of data curation, are available, such as the Global Initiative on Sharing Avian Influenza Data (GISAID) (see the website address given above). These are usually pathogen specific. Data sharing to these repositories is likely to have the collective benefit of advancing scientific understanding of the pathogen concerned [7, 8].

Data analysis

Extracting usable and actionable information from the raw output data generated by HTS instruments can be complex and time-consuming. Computational requirements often exceed those available from a standard laptop (Figure 1). Four processor cores and eight GB random access memory (RAM) may be acceptable for simple mapping of raw sequencing data to a bacterial reference genome, but anything more complex will require greater capacity. For example, with a modest computing resource (eight cores, 32 GB RAM), *de novo* assembly of a single bacterial genome can be completed within a few minutes, but extracting a viral genome from a sequenced tissue sample may take several hours (depending on the size of the initial data file). Therefore, access to high-performance computing must be a prerequisite for HTS analysis on any significant scale. This also requires disk space for storing the intermediate files generated during the analysis and significant amounts of computing memory (RAM) to ensure efficient processing of the data. The RAM required is usually scaled to the quantity of input data and complexity of the analysis. Building expertise in this area is the realm of bioinformaticians and can be challenging for scientists without training or expertise in computational biology. However, there are fundamental principles guiding the analysis for pathogen genomics that can be used as a starting point.

When the goal of HTS is to generate the whole genome sequence of a pathogen, the initial processing of raw HTS data (FASTQ files) will piece together the shorter sequencing reads generated by sequencing instruments into longer contiguous sequences (known as contigs). This is generally achieved in one of two ways:

- *de novo* assembly, in which overlaps between reads are detected and used to join them;
- mapping, which aligns reads to a matching position in a known sequence (reference).

Mapping is ideal if a quality reference genome is available, if the pathogen is known, and if the sample is not significantly different from the reference sequence. Furthermore, mapping is significantly less computationally demanding than *de novo* assembly. If the most appropriate reference is not known, a preliminary *de novo* assembly can be used to generate several contigs. Searching for the highest similarity to the contigs can then be used to select the most appropriate reference genome. High-throughput sequencing is also increasingly being used for pathogen discovery, using an untargeted metagenomics approach, and in these situations taxonomic classification of individual reads would be the most efficient approach to identifying the most likely pathogen.

The source of the data (i.e. the type of biological sample from which the HTS data are generated), and whether the pathogen is known *a priori*, are likely to be major factors in determining the analysis strategy. For example, if the source material was a pure culture of a bacterium, it is likely that direct generation of a complete genome and mapping all data to a reference sequence would be the favoured approach. Alternatively, if researchers hope to extract the genome sequence of a potentially unknown virus from a blood sample, or any sample containing non-target (host) genetic material, then pre-processing to remove unwanted sequence data would be essential.

One useful output from the mapping process is the sequence alignment map (SAM) format file, which retains the information from the input FASTQ file, together with the position and similarity to the reference genome for each of the individual sequence reads. The binary version of SAM, called 'binary alignment map (BAM), usually requires less disk space than the compressed input files so it is a useful format for longer-term data storage. Some work is being undertaken to reduce storage space requirements even further without loss of information (lossless compression). One example of this work is the compressed reference-oriented alignment map (CRAM) format [9, 10], which is widely used, especially in public repositories. However, care must be taken to ensure widespread community adoption of new formats, since this will provide a better guarantee of longevity of the software needed to process them.

De novo assembly is both more time-consuming and computationally demanding than reference mapping but is appropriate where genomic features not found in the reference genome might be expected (or in situations where a quality reference genome does not exist). Whilst the output FASTA format takes up considerably less storage space than the input data, it does not retain any quality information or traceability about its construction. Software tools and parameters used to determine whether reads are similar enough to be merged into longer contigs are still developing, so it is likely that any output could be improved upon in future iterations

of analytical software. Therefore, it is recommended that raw FASTQ files be retained to allow future re-analysis.

If removing known contamination or host sequence data is required then mapping is generally the best approach. This is because non-target reads are generally (but not in all cases) sufficiently different from the genome of the target organism and will therefore not map to the reference. Reference genomes for commonly farmed animals (e.g. cattle, pigs, chickens, etc.) are readily available from public sequence repositories (e.g. EBI, NCBI) and can be used to screen out host DNA sequences. Even when the precise species is not available, a closely related species provides sufficient similarity to remove a large proportion of host reads (e.g. the chicken reference genome is a suitable substitute for samples from turkeys). Once the unmapped reads have been collected from this process, *de novo* assembly becomes simpler and more likely to lead to characterisation of the target pathogen.

Taxonomic classification of individual reads is a useful tool for identifying pathogens in data sets without any *a priori* expectations. Individual sequence reads (or fragments of reads) are compared to databases constructed from pathogen reference genomes to determine the closest match. While these tools can process large amounts of data relatively quickly, they often require considerable computing memory. The databases are often loaded into memory to speed up data processing and this will further increase the computer hardware requirements. For example, classification with Kraken2 requires a database of reference indexes [11]. To facilitate the identification of archaea, bacteria, viruses, plasmids, or human or any other potential contamination from an HTS data set, over 50 GB RAM are required, just to store the database. Once the pathogen has been identified by this process, data can be filtered to retain the reads of interest to facilitate further characterisation of the organism, using one of the approaches listed above.

At this point, samples have been processed in isolation but, to permit comparison between samples, further analysis will be required. Phylogenetic analysis, whether based on a full genome sequence or on a 'core' genome for more variable pathogens, is often carried out to determine genetic/evolutionary relationships between samples. The standard approach is to determine the positions in the genome at which samples differ from one another (or from a reference genome), in a process known as 'variant calling'. In addition, there are standard file formats to capture this information (variant call format or VCF). The VCF output from multiple samples can then be used to generate aligned genomes, before building phylogenetic trees. At this stage, all input samples should have been processed identically to ensure robust and interpretable results. In addition, it is important to be aware that decisions made during the bioinformatics pipeline stage can bias downstream analyses so, logically, thoughts should move to the quality assurance of data management and analysis.

Quality assurance

As outputs from the analysis of HTS data are likely to lead to decisions on disease management or other animal health outcomes, there must be some assurance that the outcomes are accurate and reliable. For statutory and regulatory animal disease testing, external accreditation to laboratory testing standards (e.g. ISO 17025) is required and this should also be applied to HTS outcomes [1]. However, additional interpretation is likely to be needed (Table II).

With so many steps involved in HTS data analysis and the potential to adjust a myriad of parameters to tweak the output, it is prudent to automate data processing as much as possible, thereby ensuring reproducibility. A general requirement is to ensure that analytical processes are identical and executed sequentially in a timely fashion. There are a number of workflow managers that have been designed specifically with this in mind (e.g. Snakemake [12] or Nextflow [13]), providing a useful framework for setting up analytical pipelines. An additional advantage of using a workflow manager is that it helps to use the available computational resources efficiently.

Table II

Quality assurance of outputs from a high-throughput sequencing analysis

Requirement	Traditional testing	High-throughput sequencing analysis
Performance	Measures of sensitivity and specificity	Accuracy of detecting sequence variation
Repeatability	Is the same result achieved when a different user runs the test?	Can the analysis be repeated to generate the same outcome on a different computing device (run by a different user)?
Reproducibility	Is the same result obtained when the test is run again?	When data are reprocessed, is the outcome identical?
Traceability	Is everything used in the test (reagents, equipment, etc.) documented?	Are all software components version-controlled, and can they be linked back to the output?

Care must be taken to ensure that each of the software components has been designed and tested for the purpose for which it is used; for example, input data (sequencing instrument) and genome characteristics (e.g. high or low guanine–cytosine content bias, whether the organism is haploid or diploid, etc.). Many software tools for HTS are open source and are often continuously being developed to improve speed and/or functionality. Therefore, it is imperative that specific versions are specified and used for data analysis, and newer versions need to be thoroughly tested before embedding in an analytical pipeline.

Similarly, external data sources, such as reference genomes or taxonomic databases, may be periodically updated, so updates should always undergo fit-for-purpose testing before they are adopted. Indeed, even curated external data sources can be imperfect. Reference genomes are usually retained in public repositories (e.g. Genbank), but updated versions can be uploaded as newer, more accurate information becomes available. This may affect any outputs based on similarities to the reference genome. For example, the commonly used reference genome for *Mycobacterium bovis*, AF2122/97, was originally sequenced using traditional Sanger sequencing technology [14]. However, a revised genome with a significant number of modifications was published after the addition of HTS data from both long- and short-read technologies [15]. In these scenarios, a decision should be made as to whether data need to be re-analysed, but a note of the version of the reference genome used should always be linked to any outputs.

Accreditation bodies have only recently acquired experience in the assessment of HTS analysis. Accreditation of HTS data analysis can be achieved, but this is still an emerging area. Evidence for quality assurance activities, such as those obligatory for traditional laboratory tests, will be necessary. Interestingly, several publications have investigated ring trails, in which identical samples or data sets are sent to multiple organisations, for inter-laboratory comparisons of HTS analysis workflows [16, 17, 18]. These studies clearly indicate that many areas of HTS data generation and analysis still need to be refined. Some variability in outputs can be attributed to upstream choices of laboratory procedures and sequencing instruments. However, even when the same data sets were used as inputs, outcomes were dependent on analysis strategies, software tools and the choice of external databases.

Conclusions

Plans for the management and analysis of HTS data are often only considered after investments have been made into laboratory infrastructure. However, data storage and analysis should be a key consideration early on in the design process and will avert later difficulties as the volume and complexity of data increase. Long-term plans for data management will be essential in ensuring that outputs and analysis pipelines can be traced, and can also aid financial planning. Analysis strategies will need to be designed for each case of use, with quality assurance playing a prominent role in the maintenance and automation of data-processing pipelines. Moreover, all processes will need to remain adaptable as sequencing technology and analytical tools continue to develop.

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La gestion et l'analyse des données de séquençage à haut débit relatives aux maladies animales infectieuses

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Résumé

Les avancées technologiques dans le domaine du séquençage à haut débit (SHD) et la diminution des coûts liés à cette technique en ont accéléré l'utilisation à des fins de diagnostic et de caractérisation des maladies animales infectieuses. Le séquençage à haut débit offre plusieurs avantages par rapport aux techniques antérieures, en particulier la rapidité de son exécution et une résolution de l'ordre d'un seul changement de nucléotide parmi plusieurs échantillons, ce qui présente un grand intérêt lors des enquêtes épidémiologiques sur les foyers. Néanmoins, la pléthore de données génétiques générées en routine par le SHD devient un véritable problème en termes de stockage et d'analyse de ces données.

Les auteurs apportent un éclairage sur les aspects de la gestion et de l'analyse des données qu'il convient de prendre en compte avant d'adopter le SHD pour le diagnostic de routine en santé animale. Ces éléments relèvent de trois

catégories étroitement reliées : le stockage de données, l'analyse de données et l'assurance qualité. Chacun de ces aspects présente de nombreuses complexités et nécessitera sans doute d'être adapté à mesure que le SHD évolue. Lorsqu'elles sont prises dès la phase initiale d'un projet, des décisions stratégiques appropriées en matière d'analyse bio-informatique de séquences peuvent contribuer à éviter des problèmes majeurs sur le long terme.

Mots-clés

Assurance qualité – Bio-informatique – Caractérisation des maladies animales infectieuses – Diagnostic des maladies animales infectieuses – Données de séquençage à haut débit – Stratégie relative aux données.

Gestión y análisis de los datos de secuenciación de alto rendimiento de enfermedades animales infecciosas

R.J. Ellis & T.L. Jenkins

Resumen

Los avances tecnológicos y la reducción de los costos han acelerado el uso de la secuenciación de alto rendimiento (SAR) con fines de diagnóstico y caracterización de enfermedades animales infecciosas. La secuenciación de alto rendimiento presenta varias ventajas en comparación con otras técnicas anteriores, en particular ciclos más rápidos y una resolución que permite detectar diferencias de un solo nucleótido entre las muestras, aspectos ambos de gran importancia para el estudio epidemiológico de brotes infecciosos. Sin embargo, debido al sinnúmero de datos genéticos que constantemente se generan, no es de extrañar que esté resultando problemático almacenar y analizar los datos obtenidos.

Los autores arrojan luz sobre los aspectos de la gestión y el análisis de datos que conviene tener en cuenta antes de aplicar la SAR a las labores sistemáticas de diagnóstico en sanidad animal. Estos elementos corresponden a grandes líneas a tres categorías relacionadas entre sí: el almacenamiento de datos; el análisis de datos; y la garantía de calidad. Cada una de ellas presenta multitud de complicaciones y exige un proceso permanente de adaptación a medida que la técnica de secuenciación va evolucionando. El hecho de adoptar las buenas decisiones estratégicas sobre el análisis bioinformático de secuencias en los primeros momentos de la concepción de un proyecto ayudará a evitar importantes problemas a largo plazo.

Palabras clave

Bioinformática – Caracterización de enfermedades animales infecciosas – Datos de secuenciación de alto rendimiento – Diagnóstico de enfermedades animales infecciosas – Estrategia sobre datos – Garantía de calidad.

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